SPARSHIKA AJMAAN DINESH KUMAR

Data Engineer

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SUMMARY

Data Engineer with 4+ years of experience designing, building, and optimizing scalable batch and real-time data pipelines supporting analytics, reporting, and machine learning workloads in cloud-native environments. Proficient in Python, SQL, Apache Spark, and distributed data processing across AWS and Azure platforms. Experienced in developing ETL/ELT frameworks, streaming data pipelines, and data lake architectures that improve data availability, reliability, and performance. Strong background in performance optimization, data quality, and production-grade pipeline deployment supporting enterprise-scale data platforms.

TECHNICAL SKILLS

- **Programming & Query Languages**: Python, SQL, R
- **Data Engineering & Big Data**: Apache Spark, PySpark, Hadoop, Kafka, Airflow, Hive, Data Modeling, ETL/ELT Pipelines
- **Cloud & Data Platforms**: AWS (S3, Glue, Redshift), Azure (ADF, ADLS), Databricks
- **Databases & Warehousing**: PostgreSQL, MySQL, SQL Server, Oracle, MongoDB
- **DevOps & Workflow Automation**: Docker, Jenkins, CI/CD, Git, GitHub
- **Visualization & Analytics Support**: Power BI, Tableau, Advanced Excel

WORK EXPERIENCE

Nefroverse Technologies, USA | Data Engineer **Dec 2024 – Present**

- Designed and maintained enterprise data pipelines processing 120M+ financial transactions daily using AWS Glue, PySpark, and Redshift, improving data availability for analytics and reporting by 30%.
- Built real-time streaming data pipelines using Apache Kafka and Spark Streaming, enabling low-latency data ingestion and near real-time analytics across trading and risk systems.
- Migrated legacy on-premise ETL workflows to AWS cloud architecture, reducing infrastructure costs by $180K annually while improving system scalability and reliability.
- Optimized data processing performance through partitioning strategies, query tuning, and efficient data modeling, reducing pipeline runtime by 45%.
- Implemented data quality validation frameworks, monitoring, and automated alerting mechanisms to ensure SLA compliance and data integrity across production pipelines.
- Developed and maintained CI/CD deployment workflows using Docker, Jenkins, and Git, improving release reliability and reducing deployment time by 25%.
- Partnered with trading, risk, and analytics stakeholders to deliver production-grade data products, improving sprint predictability from 72% to 91% and accelerating reporting platform releases by 38%.

Accenture | Data Engineer (Associate Software Engineer), India **Oct 2022 - Nov 2023**

Dell EMC – iDRAC Project

- Engineered scalable telemetry and system log processing pipelines using Python and SQL to ingest, parse, and transform high-volume server diagnostic data for reliability and performance analytics.
- Designed automated data validation and Build Verification Test (BVT) frameworks using Python-based workflows, enabling structured data quality checks, anomaly detection, and regression validation across firmware release datasets.
- Developed log parsing and transformation pipelines to convert unstructured server logs into structured, analytics-ready datasets, improving failure pattern detection and downstream reporting efficiency.
- Built reusable data processing scripts and SQL-based aggregation logic to extract key performance metrics from telemetry streams, enabling faster root-cause analysis and trend monitoring.
- Implemented automated data quality validation mechanisms including schema checks, consistency validation, and threshold-based alerting to ensure integrity of diagnostic datasets.
- Optimized large-scale log analysis workflows by refactoring Python transformations and SQL queries, reducing processing time and improving telemetry analytics responsiveness.
- Enabled analytics consumption by delivering curated, structured datasets supporting reliability dashboards, performance monitoring, and operational insights across engineering teams.

Miraicoders Technology, India | Jr. Data Engineer **Jan 2021– Sep 2022**

- Collaborated in an Agile delivery environment with product, engineering, and analytics teams to improve sprint execution, increasing delivery velocity by 18% and accelerating release cycles for healthcare data platforms.
- Designed, developed, and optimized scalable ETL pipelines using Python, SQL, and Hadoop, improving end-to-end data processing throughput by 35% across high-volume clinical datasets.
- Led modernization of legacy SSIS workflows by migrating data integrations to AWS Glue and Amazon S3, reducing infrastructure overhead and enabling a fully cloud-native, scalable data integration architecture.
- Built and maintained real-time data ingestion pipelines using Apache Kafka, implementing schema validation and data quality controls that improved data accuracy and consistency by 28%
- Optimized Apache Spark batch processing jobs by refactoring transformations and tuning cluster configurations, reducing pipeline runtimes by 32% and improving data availability for analytics and machine learning workloads.

- Enabled self-service and interactive analytics by implementing Amazon Athena over partitioned S3 data lakes, reducing query latency by 22% and accelerating insights for business and clinical stakeholders.
- Implemented CI/CD pipelines for data engineering workflows using GitHub Actions and Jenkins, reducing deployment defects by 23% and improving platform reliability and operational stability.

EDUCATION

Masters in Data science
DePaul University, United States
Bachelors Industrial and Production Engineering
P.E.S. College of Engineering, India

PROJECTS

Scalable Titanic Data Pipeline Using AWS & Apache Spark

- Designed a distributed data processing pipeline on AWS EMR using Apache Spark to process 1M+ records with optimized Parquet storage.
- Improved query performance by 90% through partitioning strategies and Spark optimization techniques.
- Automated ETL workflows and enabled serverless analytics using AWS Athena for near real-time data querying.

Road Traffic Accident Data Pipeline Using AWS & Spark

- Built an end-to-end data ingestion and transformation pipeline integrating multi-source datasets into AWS S3 using Spark and AWS Lambda.
- Improved transformation throughput by 65% through Spark SQL optimization and efficient feature engineering.
- Delivered analytics-ready datasets enabling faster downstream reporting and ML model consumption.
- Enterprise Data Segmentation Pipeline with AWS Glue & Clustering
- Developed an automated AWS Glue ETL pipeline for customer segmentation using RFM analysis and PCA-based clustering.
- Reduced recurring data preparation effort by 50% through workflow automation and structured output generation.
- Delivered analytics-ready datasets for Tableau dashboards supporting business decision-making.

Machine Learning Data Pipeline for Health Risk Prediction

- Built automated data preparation and feature engineering pipelines supporting predictive modeling workflows.
- Trained and evaluated machine learning models achieving ~90% prediction accuracy using structured healthcare datasets.
- Deployed a Streamlit-based interface enabling real-time inference and interactive data exploration.

CERTIFICATIONS

- **IBM Data Engineering Professional Certificate** (2026)
- **Microsoft AI & ML Engineering Professional Certificate** (2026)
- **Advanced Data Analytics Professional Certificate** – Google (2026)
- **GenAI Data & Analytics Academy Specialization** – Starweaver (2025)
- **AWS Certified Cloud Practitioner** – In Progress